Evaluation & User Validation

Automated checks, synthetic benchmarks and early user feedback shaped the prototype.

1,217 + 132
Backend + web tests passed
Recorded run, 5 backend tests skipped

92.6%
Assistant task success
1,273 / 1,375 simulated requests

144 / 144
Safety-case executions passed
Within the same assistant simulation

30 / 30
OCR parser / table checks passed
Targeted local checks, 7 Nov 2026

01 Testing & AI Findings

Evaluation	Result	Scope / finding
Assistant stress run	92.6% task success	Simulated clinic workflows
Complex 3–5-step tasks	56 / 72 (77.8%)	Priority for further improvement
Known-failure replays	102 / 102 passed	Regression after targeted fixes
Similarity retrieval	100 / 100 Top-1	Synthetic archetype matches
Sparse reference controls	0 in Top-3	Across 100 similarity queries

Assistant: local Gemma 4 E4B. Similarity: deterministic extraction, 111 synthetic cases; each query excluded before fitting reference ranges.

02 User Feedback That Changed the Product

Cluttered website
The interface felt cluttered.
→ **UI / UX redesign**
Reorganized layouts, visual hierarchy and navigation across the clinician workspace.

Peer connection
A user wanted to talk to others at the same stage.
→ **Screened peer community**
Added tagged discussions and threaded replies with a separate public identity.

Show the OCR source
A doctor wanted to inspect the original evidence.
→ **Source-linked verification**
Added report-row crops, value highlights and correction beside the source.

03 Test Coverage & Improvements

- Workflow checks: patient access, report sharing, check-ins, consent and clinician review.
- Failure scenarios: duplicate uploads, network drops, missing devices and provider outages.
- OCR cases: results versus ranges, units, qualitative readings and Indian number formats.
- Fixes: multi-step routing, bounded timeouts, patient re-checks and observed-result verification.

Coverage combines separate documented runs and focused checks.

04 Limits & Next Validation

- Clinical quality: measure OCR field accuracy, speech errors and clinician-rated draft quality.
- User validation: document participant counts, task completion, review time and post-change feedback.
- Real-world use: test physical devices, weak networks, concurrent use, languages and accessibility.

Early feedback is qualitative. Synthetic results do not establish clinical accuracy or measured usability benefits.